

AIDED

Assisted Analysis of the Dynamic Electron Density

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1 Introduction

This is a test introduction section.

2 Calculation of the static Electron Density

The (static) Electron Density (ED) of a molecule with M positively charged nuclei and N negatively charged electrons gives the probability of finding an electron in the infinitesimally small space $d\mathbf{r}_{\mathbf{x}}$ at the point $\mathbf{r}_{\mathbf{x}} = (x, y, z)$:

$$\rho(r_x) = N \int \Psi(r_x, \mathbf{r}_2, \dots, \mathbf{r}_N; \mathbf{R}) \Psi^*(r_x, \mathbf{r}_2, \dots, \mathbf{r}_N; \mathbf{R}) d\mathbf{r}_2 \dots d\mathbf{r}_N$$

where Ψ is the spinless ground-state 'wave function' for the N electrons, distributed in the field of nuclei with fixed positions (given by the 3M vectors $\mathbf{R}$).

Given an equilibrium geometry $\mathbf{R}$ and accompanying ground-state wave function Ψ , a common practice is to describe the many electron wavefunction, Ψ_i , as an antisymmetric linear combination of one-electron Molecular Orbitals (MOs), Ψ_i , each of which is described by a linear combination of Atomic Orbitals (AO), ϕ_j .

An atomic orbital (AO) is a contraction of primitive Gaussians:

$$\phi_j(\mathbf{r}) = \sum_k g_{jk} g_k(\mathbf{r}) \quad (1)$$

Molecular orbitals (MOs) are written directly in the primitive basis using the coefficients that appear in the code:

$$\Psi_i(\mathbf{r}) = \sum_k C_{ik} g_k(\mathbf{r}) \quad (2)$$

The electron density is

$$\rho(\mathbf{r}) = \sum_{\mu, \nu} P_{\mu\nu} \phi_{\mu}(\mathbf{r}) \phi_{\nu}(\mathbf{r}), \quad P_{\mu\nu} = 2 \sum_i n_i c_{i\mu} c_{i\nu} \quad (3)$$

(AO-level MO coefficients $c_{i\mu}$ are related to C_{ik} by $C_{ik} = \sum_{\mu} c_{i\mu} g_{\mu k}$.)

In AIDED we insert (1) into (3):

$$\begin{aligned}\rho(\mathbf{r}) &= \sum_{\mu,\nu} P_{\mu\nu} \left(\sum_k g_{\mu k} g_k(\mathbf{r}) \right) \left(\sum_l g_{\nu l} g_l(\mathbf{r}) \right) \\ &= \sum_{k,l} \left(\sum_{\mu,\nu} g_{\mu k} P_{\mu\nu} g_{\nu l} \right) g_k(\mathbf{r}) g_l(\mathbf{r}).\end{aligned}\quad (4)$$

Define the primitive-basis density matrix

$$D_{kl} = \sum_{\mu,\nu} g_{\mu k} P_{\mu\nu} g_{\nu l}, \quad (5)$$

so that

$$\rho(\mathbf{r}) = \sum_{k,l} D_{kl} g_k(\mathbf{r}) g_l(\mathbf{r}) = \mathbf{g}^T D \mathbf{g}.$$

Because Gaussian's *.wfn file already folds the contraction coefficients into the exported MO matrix C_{ik} , we may build D in a *single pass* exactly as in the Python routine:

$$D_{kl} = \sum_i n_i C_{ik} C_{il}. \quad (6)$$

The entire grid calculation then reduces to evaluating $\rho(\mathbf{r}) = \mathbf{g}^T D \mathbf{g}$ with a **fixed** D , giving the computational speed-up described in the text.

3 MSDA: Mean Square Displacement Amplitudes

The normal mode analysis of a system containing N vibrating nuclei in thermal equilibrium leads to a $3N$ -multivariate normal distributino of nuclear displacements relative to their equilibrium positions ($\mathbf{u} = \mathbf{R} - \mathbf{R}^0$):

$$P(\mathbf{u}) = (2\pi)^{-3N/2} |\mathbf{U}|^{-1/2} \exp \left(-\frac{1}{2} \mathbf{u} \mathbf{U}^{-1} \mathbf{u}^T \right) \quad (7)$$

where $\mathbf{u}^T$ represents the transpose of $\mathbf{u}$ and $\mathbf{U}$ is the covariance matrix of the displacements.

This covariance matrix is the Mean Square Displacement Amplitude (MSDA) matrix associated with the expectation values of the Cartesian nuclear displacement products (second moments):

$$\mathbf{U} = \langle \mathbf{u} \mathbf{u}^T \rangle_T$$

The temperature dependence of $\mathbf{U}$ is embedded in the MSDAs of the normal modes in its eigenvalues $\delta = (\delta_{i=1,3N})$:

$$\mathbf{U} = \mathbf{L} \delta \mathbf{L}^T \quad (8)$$

where $\mathbf{L}$ is the matrix of eigenvectors (normal modes) of the Hessian matrix.

$$\delta_j = \frac{h}{8\pi^2\nu_j} \coth\left(\frac{h\nu_j}{2k_B T}\right) \quad (9)$$

where ν_j is the frequency of the j -th normal mode, h is Planck's constant, and k_B is Boltzmann's constant.

3.1 MSDA implementation in AIDED

In the `aided.io.vib.gaussian` module the Eigen-representation (9) is evaluated by the following fragment `gaussian.py::GaussianLogReader.gen_msda`:

```
delta = ZPE_PREF_ANG2 / freqs / reduced_masses #  $\delta_j$  in  $\text{\AA}^2$ 
if temperature > 1e-9:
    mode_prefactor *= coth(THERM_FACTOR_CM_K * freqs / temperature)
    msda = np.einsum("is,s,js->ij", modes, delta, modes)
```

- `ZPE_PREF_ANG2` implements $h/(8\pi^2c)$ together with unit conversions ($\text{cm}^{-1} \rightarrow \text{s}^{-1}$ and $\text{amu} \rightarrow \text{\AA}^2$), so that the resulting δ_j are directly in $\text{\AA}^2$.
- `THERM_FACTOR_CM_K` := $hc/(2k_B)$ converts the argument of the `coth` to dimensionless form.
- The division by `reduced_masses` in the prefactor converts the stored *mass-weighted* eigenvectors `modes` into the Cartesian $\mathbf{L}$ used in Eq. (9). Omitting this term would yield the mass-weighted MSDA.
- The Einstein summation $U_{ij} = \text{modes}(is)\delta_s\text{modes}(js)$ realises $\mathbf{U} = \mathbf{L}\delta\mathbf{L}^T$ efficiently.

Consequently, the MSDA matrix produced by the code is numerically identical to Eq. (9) and all elements are expressed in Cartesian coordinates with units of $\text{\AA}^2$.

4 The dynamic Electron Density

Given (7) we can define the dynamic electron density as

$$\langle \rho(\mathbf{r}) \rangle_T = \sum_a \int_{-\infty}^{\infty} \rho_a(\mathbf{r}_a - \mathbf{u}_a) P_a(\mathbf{u}) d\mathbf{u} \quad (10)$$

where

$$P_j(\mathbf{u}) = (2\pi)^{-3/2} |\mathbf{U}_j|^{-1/2} \exp\left(-\frac{1}{2} \mathbf{u} \mathbf{U}_j^{-1} \mathbf{u}^T\right) \quad (11)$$

where $\mathbf{u}_j$ is the displacement of the j^{th} nucleus from its equilibrium position, and $\mathbf{U}_j$ is the symmetric 3x3 tensor which represents the variance-covariance of

the nuclear PDF. The Anisotropic Displacement Parameters (ADP) which make up the tensor $\mathbf{U}_j$ represent the likelihood for the j^{th} nucleus to be displaced in given directions. These symmetric 3X3 matrices are the variance-covariance in $\mathbf{u}_j = (x_j, y_j, z_j)$ for nucleus j .

The block diagonal elements of the Cartesina MSDA represent the ADPs for each nucleus and can be used to define probability ellipsoids.

4.1 equations

Atom-resolved dynamic electron density

Definition.

$$\rho_{\text{dyn},i}(\mathbf{r}) = \int_{\mathbf{R}^3} \rho_{\text{stat},i}(\mathbf{r} - \mathbf{s}) p_i(\mathbf{s}) d^3s, \quad p_i(\mathbf{s}) = \frac{\exp[-\frac{1}{2} \mathbf{s}^T \mathbf{U}_{ii}^{-1} \mathbf{s}]}{(2\pi)^{3/2} \sqrt{\det \mathbf{U}_{ii}}}. \quad (1)$$

Static atomic density in a contracted Gaussian basis.

$$\rho_{\text{stat},i}(\mathbf{r}) = \sum_{\substack{\mu \in i \\ \nu}} P_{\mu\nu} g_{\mu}(\mathbf{r}) g_{\nu}(\mathbf{r}), \quad g_{\mu}(\mathbf{r}) = N_{\mu} (\mathbf{r} - \mathbf{A})^{\mathbf{n}_{\mu}} e^{-\alpha_{\mu} \|\mathbf{r} - \mathbf{A}\|^2}. \quad (2)$$

Effective covariance for one primitive pair. With doubled exponents $\alpha = 2\alpha_{\mu}$, $\beta = 2\alpha_{\nu}$, $\gamma = \alpha + \beta$ and centres $\mathbf{A}, \mathbf{B}$, the harmonic convolution yields

$$\boxed{\Sigma_{ij} = \frac{\mathbf{I}}{2\gamma} + \frac{\alpha^2 \mathbf{U}_{ii} + \beta^2 \mathbf{U}_{jj} + \alpha\beta(\mathbf{U}_{ij} + \mathbf{U}_{ij}^T)}{\gamma^2}, \quad \Lambda_{ij} = \Sigma_{ij}^{-1}} \quad (3)$$

Thermally smeared primitive product. Let $\mathbf{C} = \frac{\alpha\mathbf{A} + \beta\mathbf{B}}{\gamma}$

Then for s -type primitives

$$[g_{\mu} g_{\nu}]_{\text{dyn}}(\mathbf{r}) = N_{\mu\nu} \exp\left[-\frac{\alpha\beta}{\gamma} \|\mathbf{A} - \mathbf{B}\|^2\right] \exp\left[-\frac{1}{2}(\mathbf{r} - \mathbf{C})^T \Lambda_{ij}(\mathbf{r} - \mathbf{C})\right], \quad (4)$$

where $N_{\mu\nu} = N_{\mu} N_{\nu} (\det \Sigma_{ij})^{-1/2}$.

Including Cartesian angular momenta. TODO: Show how to include angular momenta in the dynamic density.